

Question D5:

Pseudocode:

START

DECLARE N, i, capA, capB, capC, occA, occB, occC, processed, accepted,
Rount, Cparked, Bparked, Vparked, spaces: INTEGER

DECLARE vehicle, user, permit, Target2: CHAR rejectR, Hzone: STRING

DECLARE valid, assigned: BOOLEAN

capA $\leftarrow$ 20, capB $\leftarrow$ 40, capC $\leftarrow$ 15, occA $\leftarrow$ 0, occB $\leftarrow$ 0,

occC $\leftarrow$ 0, processed $\leftarrow$ 0, accepted $\leftarrow$ 0, rejected $\leftarrow$ 0,

Cparked $\leftarrow$ 0, Bparked $\leftarrow$ 0, Vparked $\leftarrow$ 0.

INPUT N

FOR i $\leftarrow$ 1 to Nvalid $\leftarrow$ FALSEemergency $\leftarrow$ 'N'

WHILE valid = FALSE

OUTPUT "Enter vehicle type (C/B/V): "

INPUT vehicle

OUTPUT "Enter user category (F/S/V): "

INPUT user

OUTPUT "Enter Permit Validity (Y/N): "

INPUT permit

IF permit = 'N' THEN

OUTPUT "Is it an emergency? (Y/N): "

INPUT emergency

ENDIF

IF (vehicle = 'C' OR vehicle = 'B' OR vehicle = 'V') AND (user = 'F'
OR user = 'S' OR user = 'V') AND (permit = 'Y' OR
permit = 'N') THEN

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valid = TRUE

ENDIF

ENDWHILE

processed $\leftarrow$ processed + 1

assigned $\leftarrow$ FALSE

spaces $\leftarrow$ 1

IF permit = 'N' AND emergency = 'N' THEN

OUTPUT "Invalid permit and no emergency"

ELSE

IF user = 'F' THEN

Target2 $\leftarrow$ 'A'

spaces $\leftarrow$ 1

ELSE IF user = 'S' THEN

IF vehicle = 'V' THEN

Target2 $\leftarrow$ 'C'

spaces $\leftarrow$ 2

ELSE

Target2 $\leftarrow$ 'B'

spaces $\leftarrow$ 1

ENDIF

ELSE IF user = 'G' THEN

Target2 = 'C'

IF vehicle = 'V' THEN

spaces $\leftarrow$ 2

ELSE

spaces $\leftarrow$ 1

ENDIF ENDIF

```
IF Target2 = 'A' THEN
```

```
  IF (capA - occA) >= spaces THEN
```

```
    occA ← occA + spaces
```

```
    assigned ← TRUE
```

```
    rejectR ← "Assigned to Zone A. Remaining Capacity =  
              (capA - occA)."
```

```
  ELSE
```

```
    OUTPUT "No available space in Zone A"
```

```
  ENDIF
```

```
ELSEIF Target2 = 'B' THEN
```

```
  IF (capB - occB) >= spaces THEN
```

```
    occB ← occB + spaces
```

```
    assigned ← TRUE
```

```
    OUTPUT "Assigned to Zone B. Remaining capacity =  
           (capB - occB)"
```

```
  ELSE
```

```
    rejectR ← "No available space in Zone B."
```

```
  ENDIF
```

```
ELSEIF Target2 = 'C' THEN
```

```
  IF (capC - occC) >= spaces THEN
```

```
    occC ← occC + spaces
```

```
    assigned ← TRUE
```

```
    OUTPUT "Assigned to Zone C. Remaining capacity =  
           (capC - occC)"
```

```
  ELSE
```

```
    rejectR ← "No available space or suitable zone"
```

```
  ENDIF
```

```
ENDIF
```

```
ENDIF
```

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IF assigned = 'TRUE' THEN

accepted $\leftarrow$ accepted + 1

IF vehicle = 'C' THEN

Cparked $\leftarrow$ Cparked + 1

ENDIF

IF vehicle = 'B' THEN

Bparked $\leftarrow$ Bparked + 1

ENDIF

IF vehicle = 'V' THEN

Vparked $\leftarrow$ Vparked + 1

ENDIF

ELSE

Rcount $\leftarrow$ Rcount + 1

OUTPUT "Vehicle rejected. Reason: " reject R

ENDIF

NEXT

OUTPUT "Summary:"

OUTPUT "Total vehicles processed: " processed

OUTPUT "Total Accepted: " accepted " | Total Rejected: " Rcount

OUTPUT "Cars Parked: " Cparked " | Bikes: " Bparked " | Vans: " Vparked

OUTPUT "Zone A $\Rightarrow$ Final Occupancy: " occA " | Remaining Capacity: "
(capA - occA).

OUTPUT "Zone B $\Rightarrow$ Final Occupancy: " occB " | Remaining Capacity: "
(capB - occB).

OUTPUT "Zone C $\Rightarrow$ Final Occupancy: " occC " | Remaining Capacity: "
(capC - occC).

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IF $occA \geq occB$ AND $occA > occC$ THEN

Hzone $\leftarrow$ "Zone A"

ELSE IF $occB \geq occA$ AND $occB \geq occC$ THEN

Hzone $\leftarrow$ "Zone B"

ELSE

Hzone $\leftarrow$ "Zone C"

ENDIF

OUTPUT "Zone with highest occupancy: " Hzone

IF $(occA + occB + occC) = (capA + capB + capC)$ THEN

OUTPUT "The entire campus parking facility is full"

ELSE

OUTPUT "The campus parking facility is not full."

ENDIF

END

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PAC

Data Given (INPUT)	Processing required	Required Results
→ Total Vehicles: N → Vehicle type: vehicle → User category: user → Permit status: permit → Emergency status: emergency	(1) Execute a loop N times to gather inputs (2) Evaluate target zones and space constraints based on user/vehicle types. (3) Perform available capacity checks for Zones A, B, C prior to assignment. (4) Increment accept/reject totals and tally parked vehicle types (5) Compare final occupancies to identify the most utilized parking zones. (6) Evaluate if total occupancies matches total capacity to check if the campus is full	- Real time zone assignment or rejection messages. - End of process summary statistics for processed, accepted, and rejected totals. - Final occupancy and capacity metrics per zone - Highest occupancy zone declaration - Campus facility full/not full status message

Alternative Rules:-

(i) If permit = 'N' and emergency = 'N', the vehicle request is invalid.

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IPO

INPUT

- N
- Vehicle
- user
- permit
- emergency

Process

- Initialize zones, capA=20, capB=40, capC=15 and the tracker to zero.
- loop N times to input and validate vehicle, user and permit data
- Reject the vehicle immediately if permit is 'N' and emergency is 'N'.
- Determine the target zones
- Check if the designated target zone has enough remaining capacity for required spaces
- Check capacity permits, update occupancies or invalidate
- Calculate the zone with the highest occupancy.
- Determine if the total confirmed occupancy equals the total campus capacity.

OUTPUT

- Assignment confirmation or rejection reason
- Total vehicles processed, total accepted, total rejected
- Total counts of Cars, Bikes, and vans parked.
- Final occupancy and remaining capacity.
- Zones with the highest occupancy,
- Message stating whether the overall campus parking facility is full.

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STAR

capA ← 20, capB ← 40, capC ← 15
occA ← 0, occB ← 0, occC ← 0
processed ← 0, accepted ← 0, Cparked ← 0
Bparked ← 0, Vparked ← 0

INPUT N

ic ← 1

Yes No
is
ic = N

valid ← FALSE

Emergency ← 'N'

valid = FALSE?

No

processed ←
processed + 1
assigned ← FALSE
spaces ← 1

Yes permit = 'N' AND
Emergency = 'N'

No

OUTPUT
"Invalid permit
and no
emergency"

user = 'F'?

No

B

OUTPUT "summary"

OUTPUT "Total vehicles processed:" processed
OUTPUT "Total accepted:" accepted "Total
rejected:" Rejected
OUTPUT "Cars Parked:" Cparked "Bikes:"
Bparked "Vans parked:" Vparked

OUTPUT "Zone A: Final occupancy = " occA

"final capacity = " capA - occA

OUTPUT "Zone B: Final occupancy = " occB

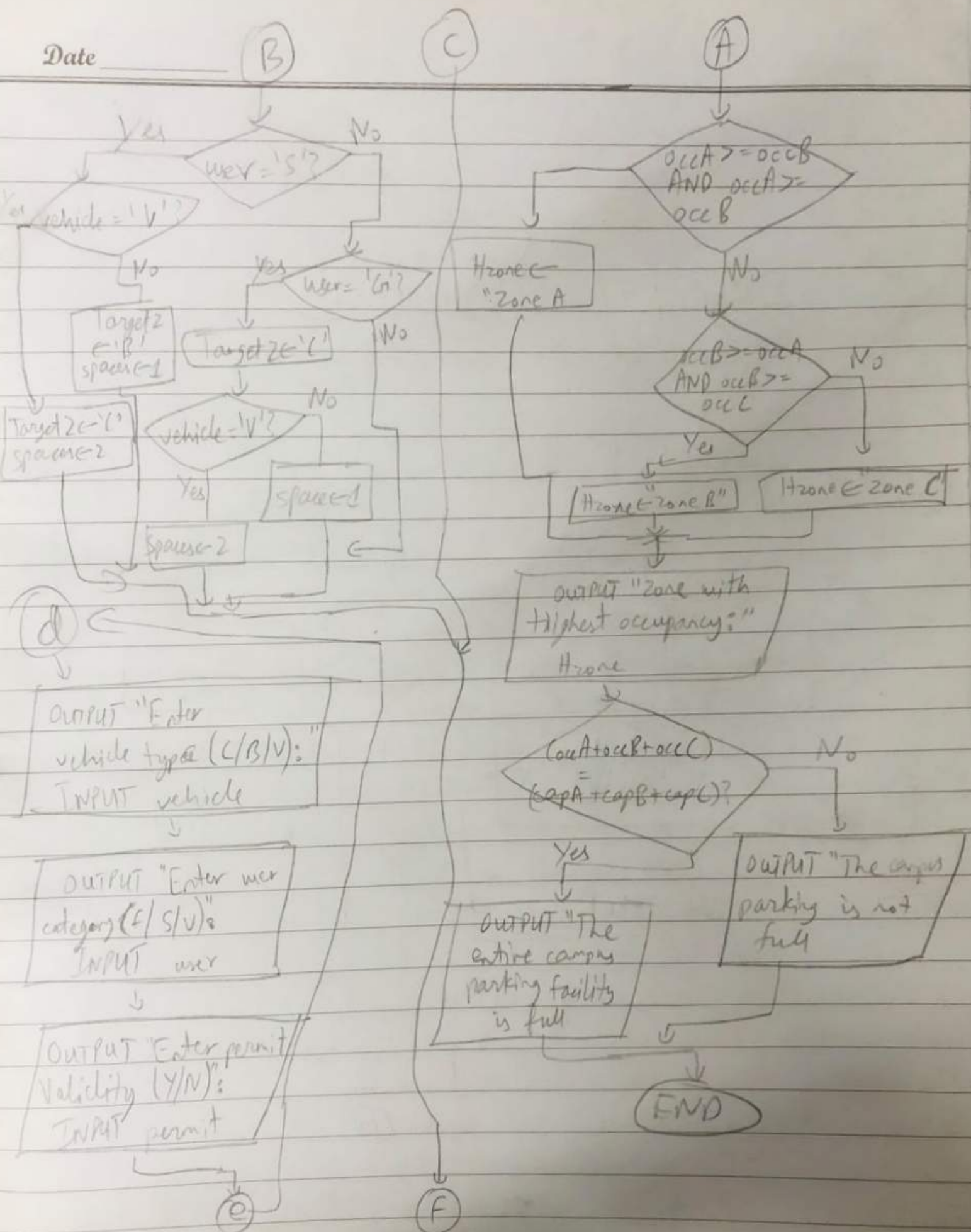
"final capacity = " capB - occB

OUTPUT "Zone C: Final occupancy = " occC

"final capacity = " capC - occC

A

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Date _____

